

An isometric illustration of a city skyline with various skyscrapers in shades of blue and teal. Some buildings have unique features like a helipad, a dollar sign on top, a Wi-Fi symbol, a padlock, a helicopter, and a checkmark. The background is a solid dark blue.

NEW YORK CITY FOR-HIRE VEHICLES VS PUBLIC TRANSIT

CIS 5450 Big Data Analytics
Final Project

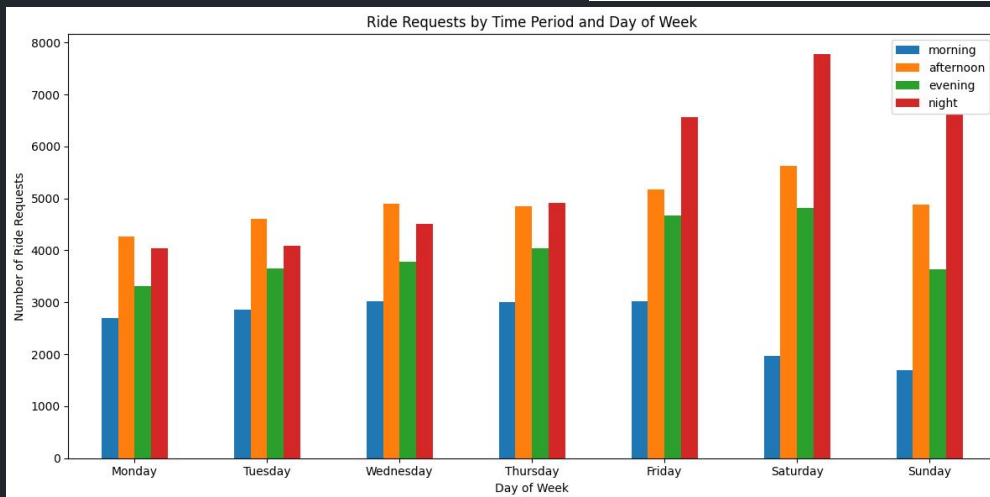
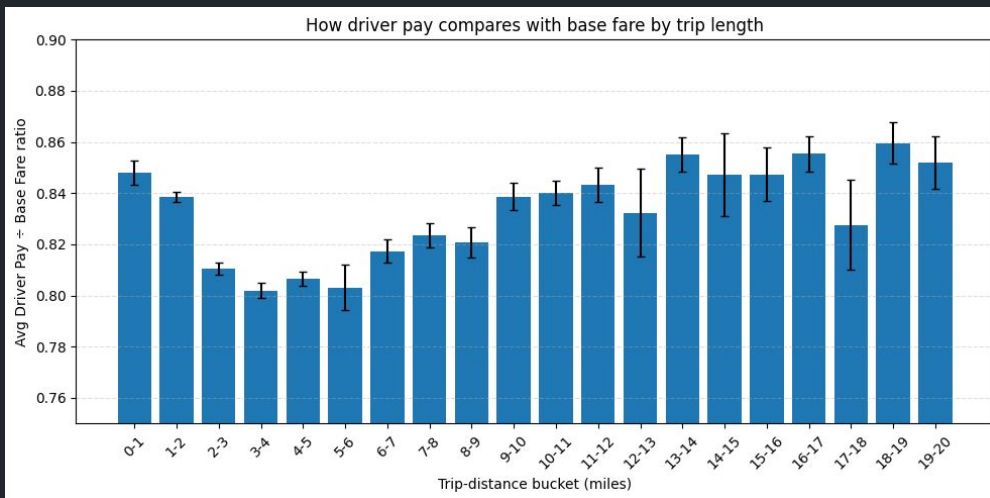
OBJECTIVE + VALUE
PROPOSITION

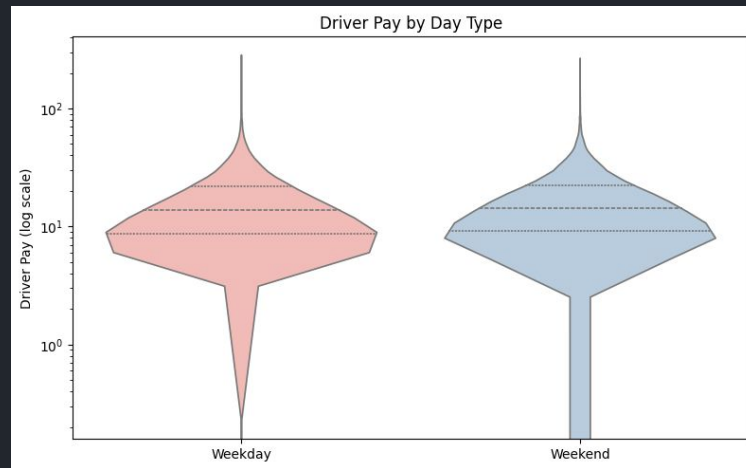
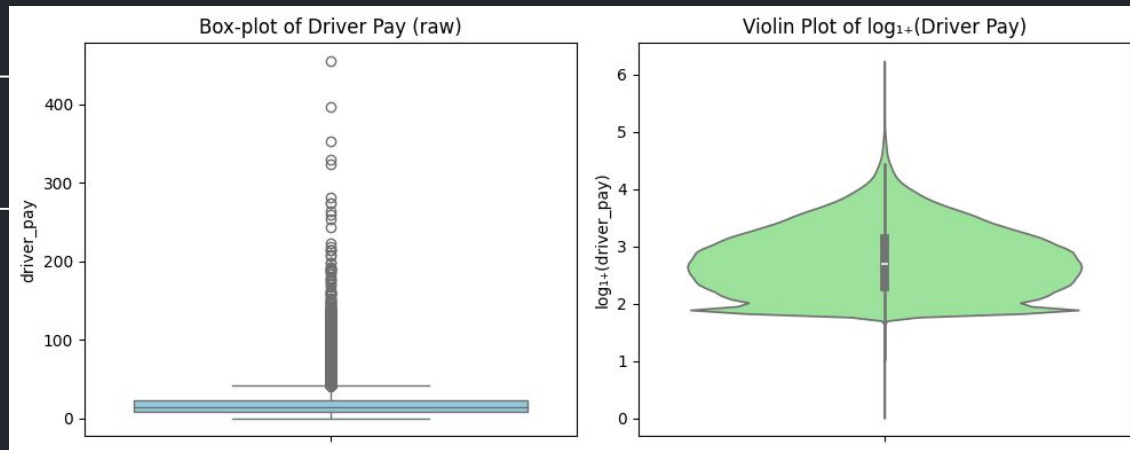
- For-Hire-Vehicles (FHVs) in NYC have unpredictable schedules, incomes, and service patterns.
- Lack of transparent algorithms for Uber and Lyft
- NYC Taxi and Limo Commission (TLC) dealing with frequent outrage
- Analyze data on payment structures, trip durations, and pickup zones to build predictive tools
- Empower drivers and rides for equal pay and efficiency

DATASET USED

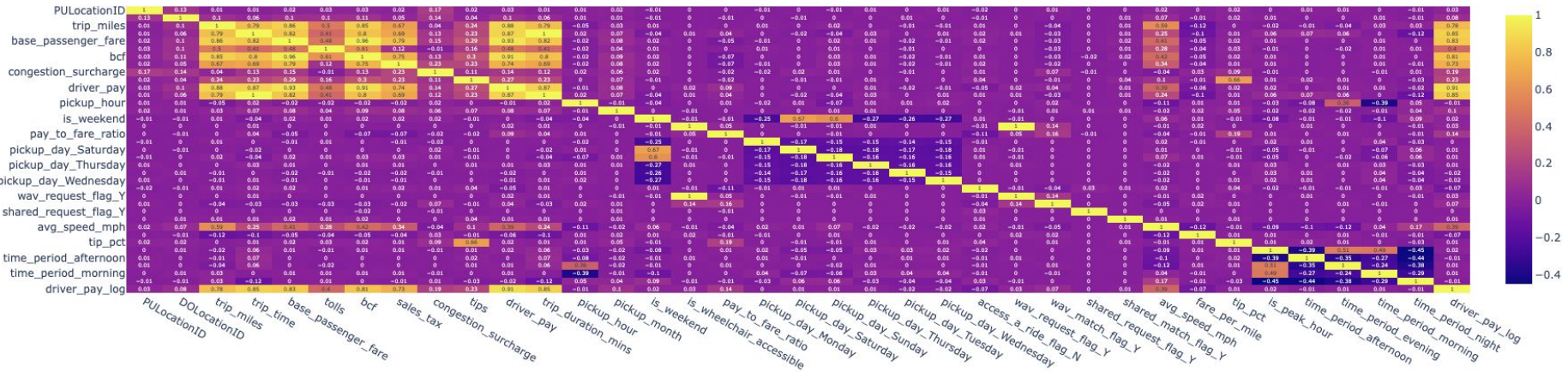
- FHV Data about NYC in 2021
- Random sampling of data given large size
- Cleaned and engineered features:
 - Day of the week
 - Month of trip
 - Pay-to-Fare ratio
 - Weekend

Major Learnings from eda

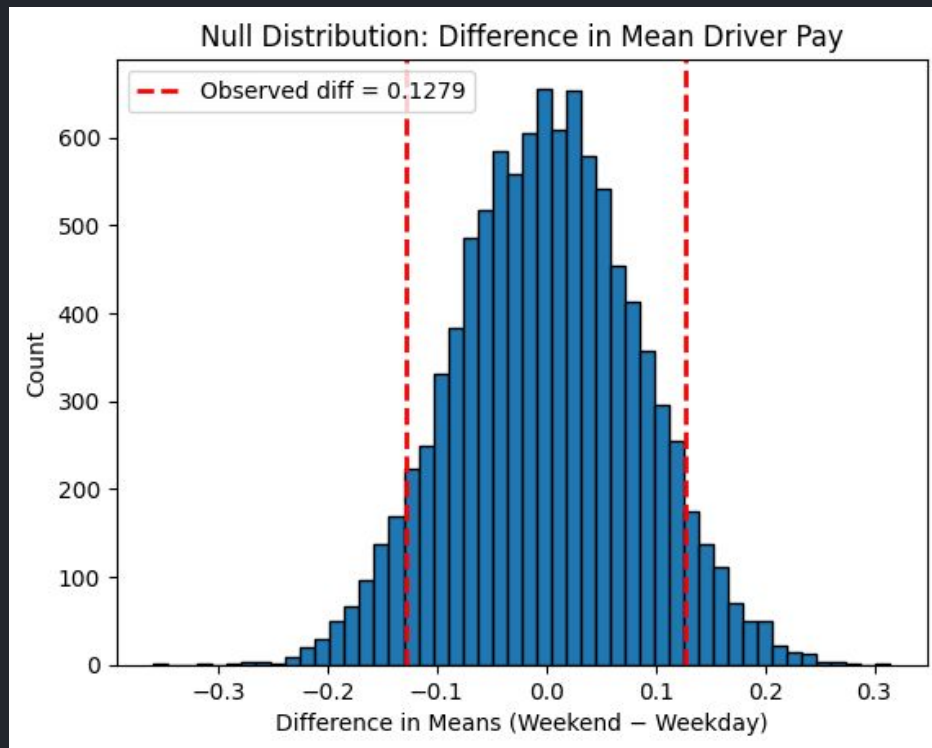




Feature Correlation Matrix (including driver_pay)

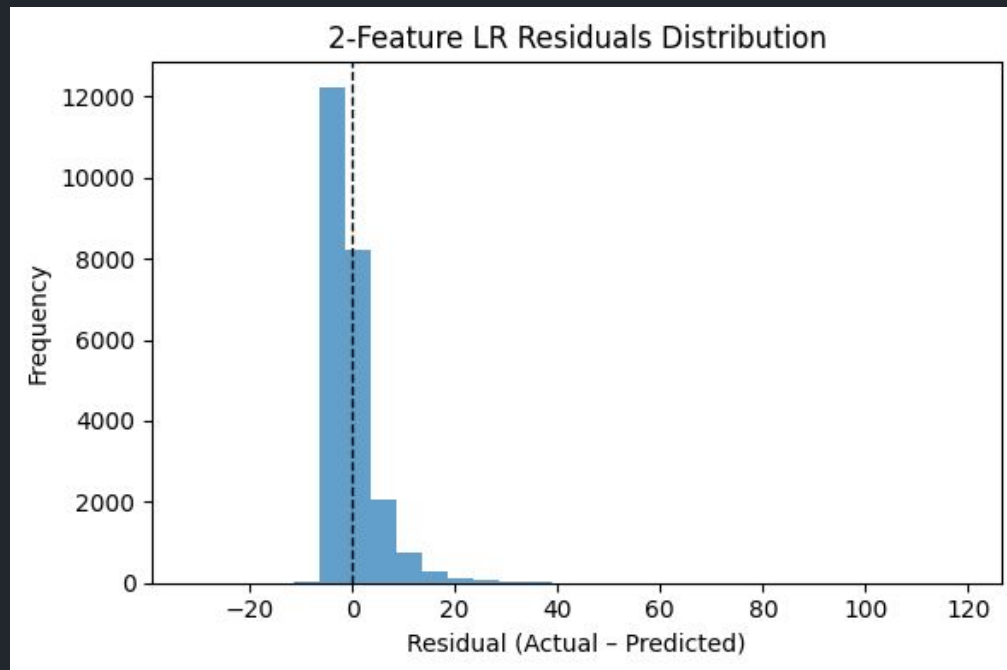
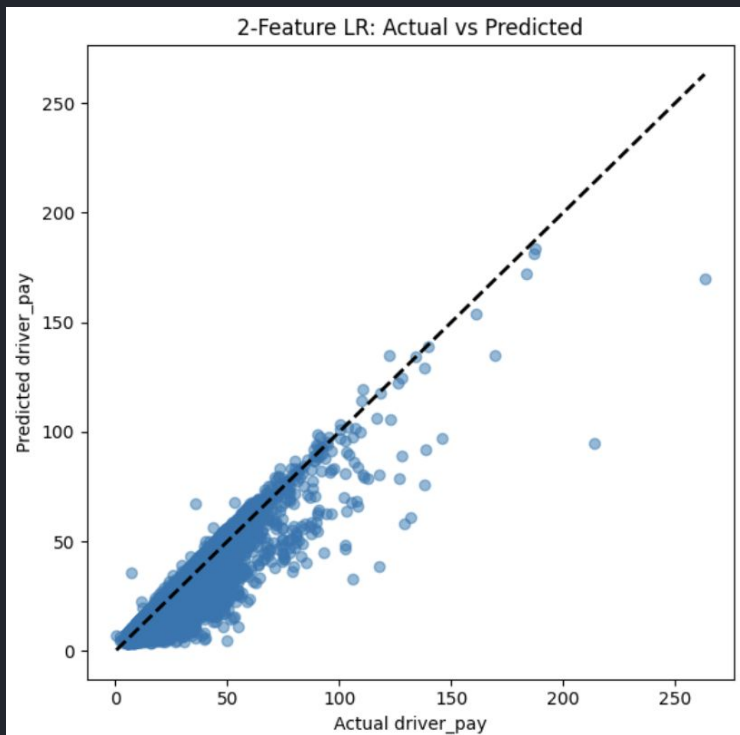


HYPOTHESIS TEST RESULTS



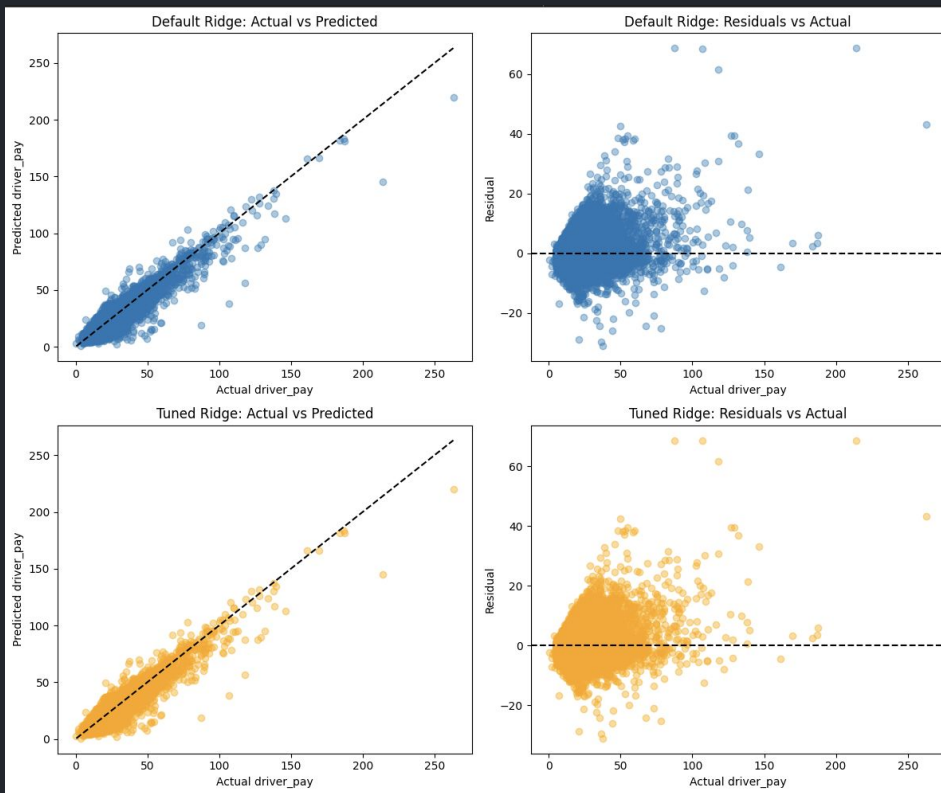
MODELING RESULTS

Multiple Linear Regression Baseline



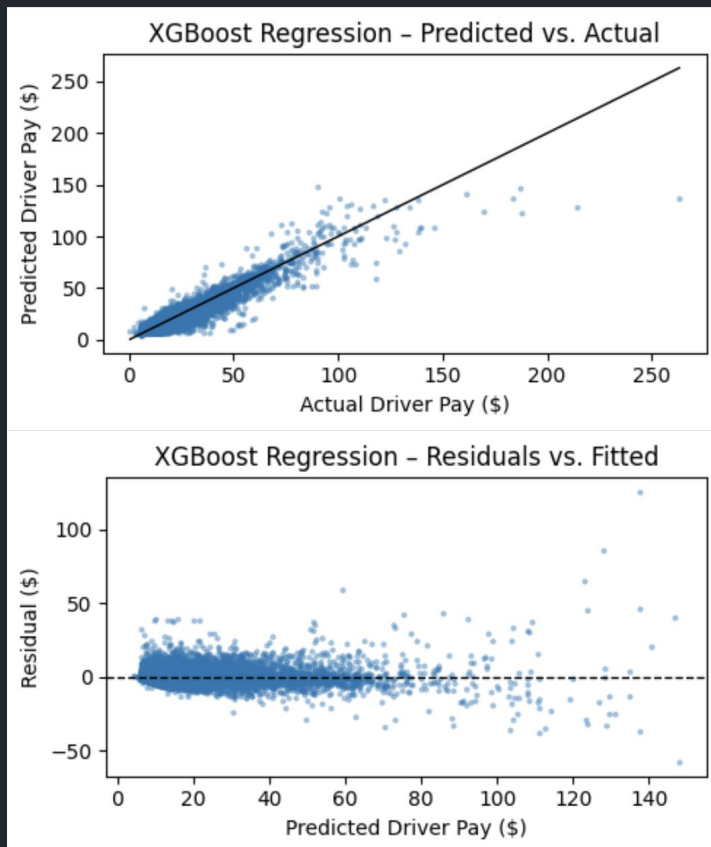
MODELING RESULTS

Ridge Regression



MODELING RESULTS

XGBoost Regression

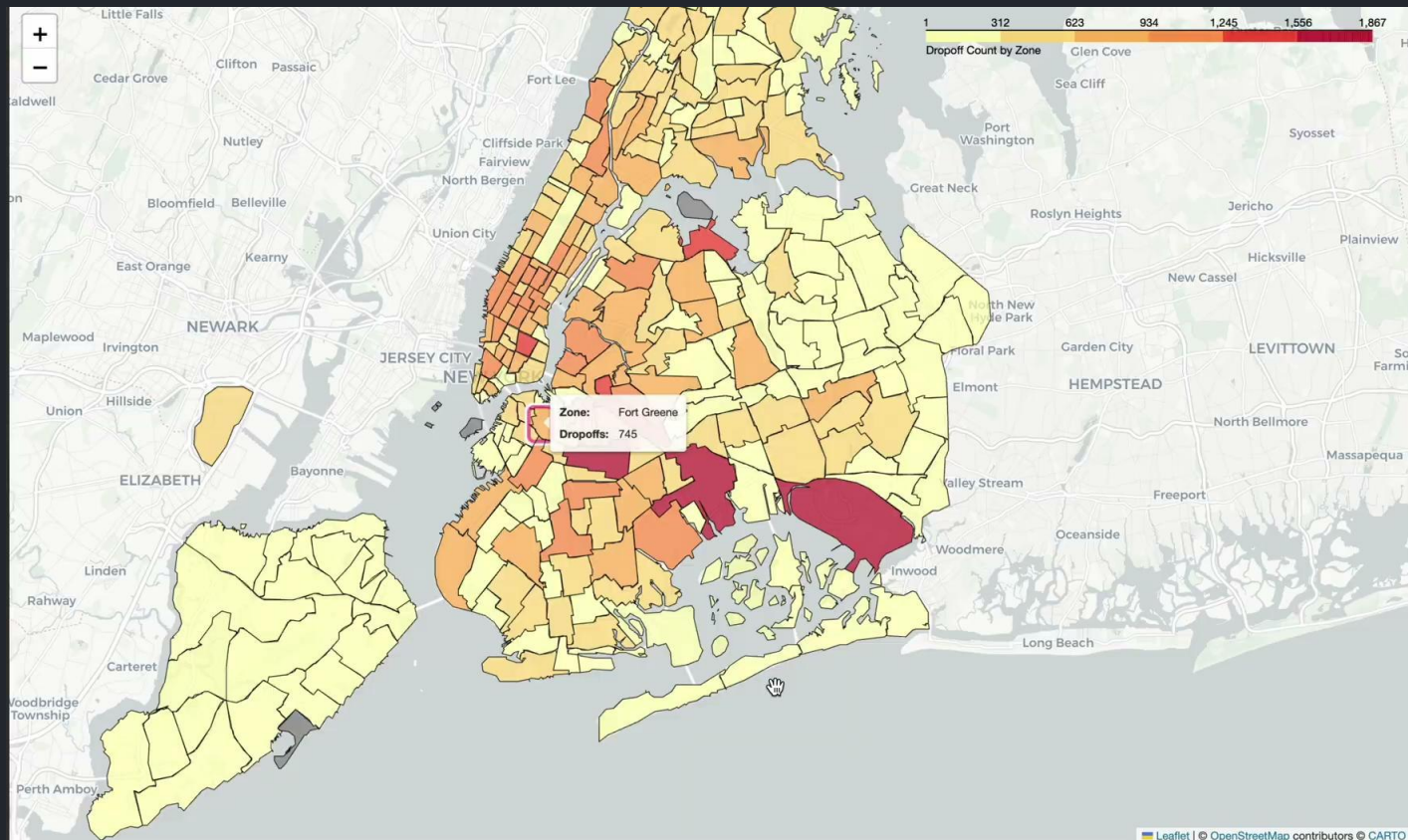


MODELING RESULTS

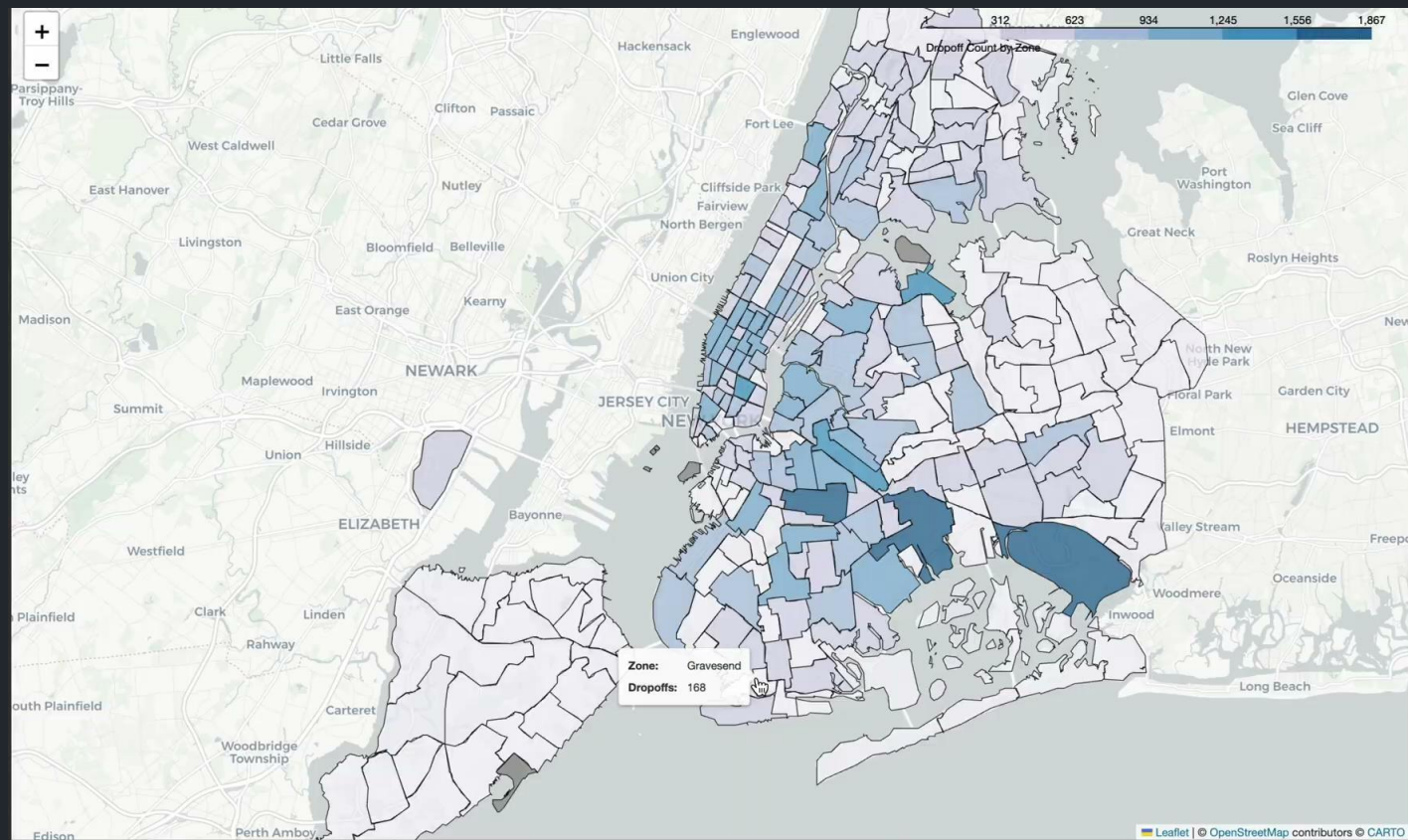
MODEL ASSESSMENT

	Model	Training R ²	Testing R ²	Training RMSE	Testing RMSE	Training MAE	Testing MAE
0	2-Feature LR Baseline	0.859903	0.859677	4.973591	4.965444	3.167235	3.163314
1	Default Ridge	0.912438	0.911952	3.931990	3.933267	2.509106	2.513772
2	Tuned Ridge	0.912438	0.911952	3.931990	3.933252	2.509158	2.513816
3	Default XGBoost	0.934517	0.910373	3.400320	3.968377	2.142151	2.325801
4	Tuned XGBoost	0.919940	0.911628	3.759784	3.940495	2.274212	2.330694

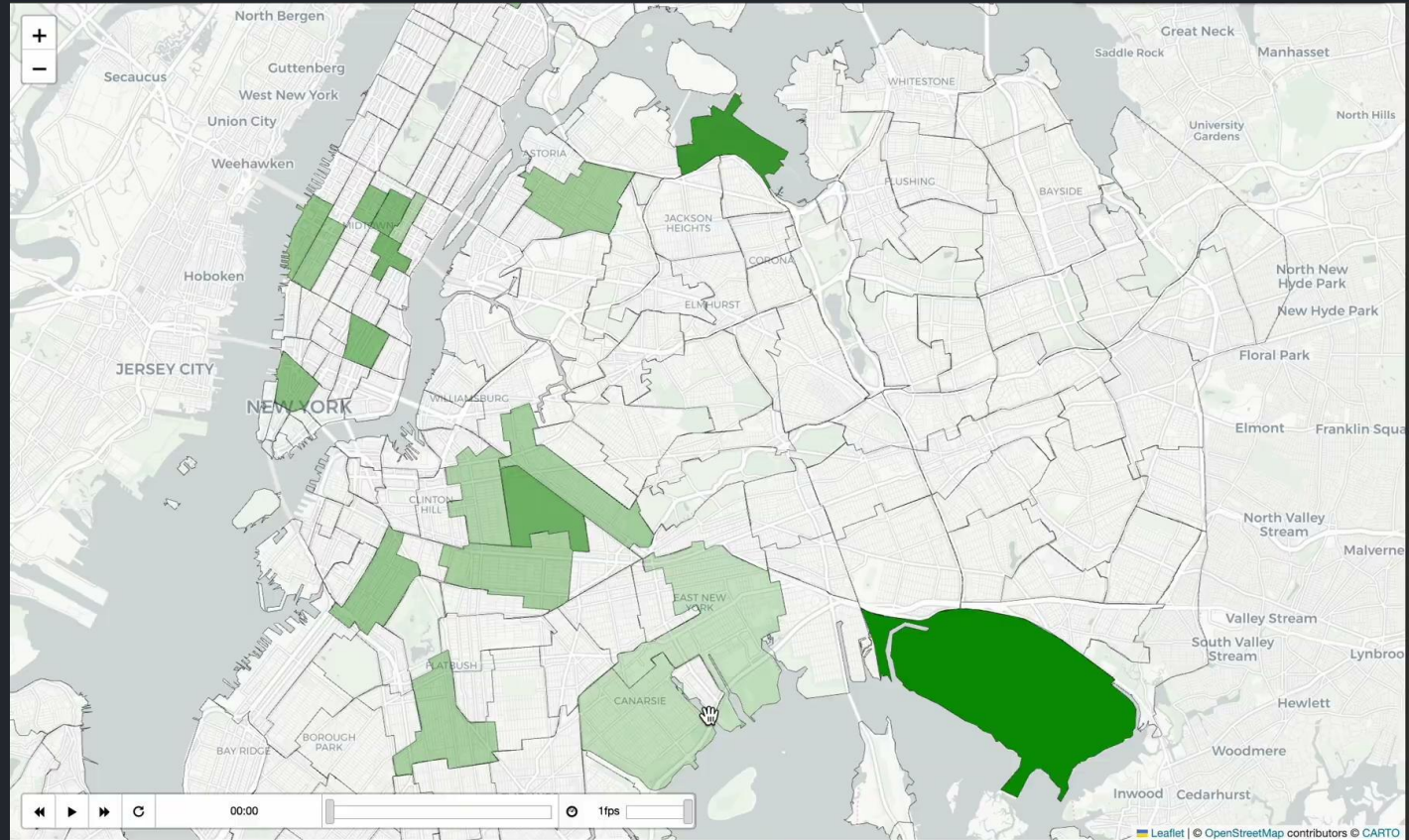
MAPPING OF DROP OFF POINTS



MAPPING OF PICKUP POINTS

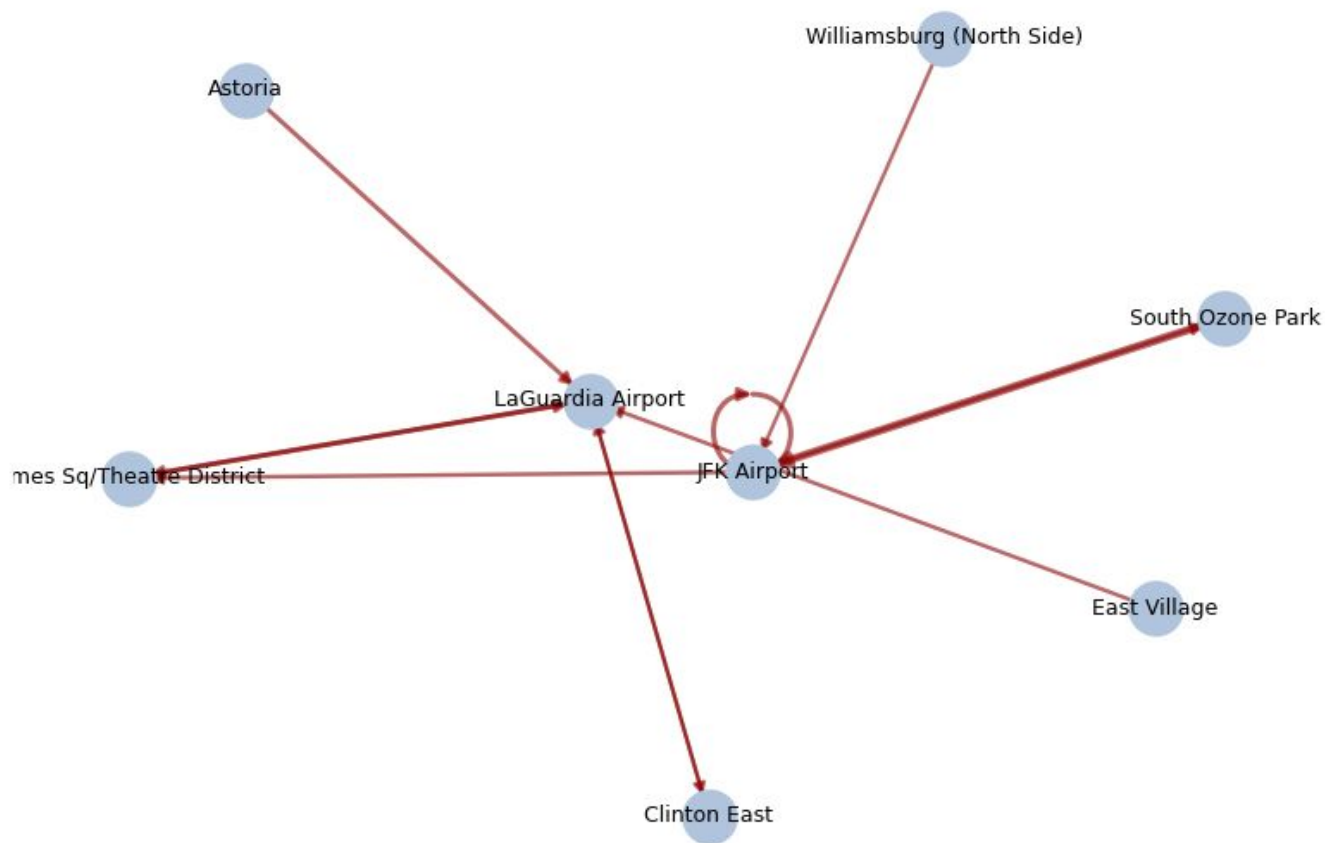


MAPPING OF DRIVER PAY VS time



NETWORK OF AIRPORT PICKUPS

Airport-Linked Trips: PU/DO Zones Matching JFK, LGA, EWR



IMPLICATIONS AND INSIGHTS

- EDA:
 - Pay varies more when trip length is longer
 - Airport-based rides make up large proportion of FHV rides
 - Some locations across the city (Astoria, Williamsburg, East Village) most popular for airport pickups -> improved infrastructure for airport transit
- Modeling:
 - Trip duration, trip distance, and tip amount are largest determinants
 - Heteroskedasticity: more variance with higher-pay rides
 - Congestion surcharge, weekend flag, wheelchair-accessibility are useful for prediction
 - How can drivers predict their take-home pay?

CHALLENGES,
LIMITATIONS,
FUTURE

- Data granularity
 - Ex- insights about total pay earned by each driver over time
- Algorithmic opaqueness
- Post-pandemic surge pricing given 2021 data available
- Pay versus profit
 - What are driver's costs? Licensing, gas
 - Does high pay zone = high profit?
- Changes in TLC policy (minimum wages)